Prior-adaptive evidence thresholds for clinical variant classification: limitations of the Tavtigian framework and a piecewise Bayesian solution

Abstract

Evidence-based variant classification under the ACMG/AMP guidelines requires mapping continuous likelihood-ratio scores to discrete evidence strengths using thresholds that satisfy specified posterior probability targets. The Tavtigian et al. framework provides such thresholds through a single constant C^* calibrated to a canonical prior of $p = 0.10$, but the mapping is exact only at that prior and degrades systematically at all others. We characterise the failure modes of this approach in detail: pathogenic thresholds drift from their targets at priors above ∼0.20, while benign thresholds are severely miscalibrated across the entire biologically relevant range—at $p = 0.01$, the LB boundary posterior is 0.003 instead of 0.10, causing benign evidence to be withheld from the vast majority of variants that should receive it. We prove that no additive scoring system with prior-independent thresholds can simultaneously produce exact posteriors at all classification boundaries for all priors, and we show that a piecewise-linear Bayesian solution achieves exact boundary posteriors at every prior while preserving the ACMG/AMP point system. We demonstrate that the threshold difference is practically large: at $p = 0.01$, the piecewise LB threshold ($\mathrm{LR}^+{<}11$) is 42× more permissive than the Tavtigian threshold ($\mathrm{LR}^+{<}0.26$), directly reducing false-negative benign evidence assignments in assay calibration pipelines.

Introduction

The ACMG/AMP guidelines [1] classify genetic variants into five categories according to their posterior probability of pathogenicity: pathogenic (≥99%), likely pathogenic (LP; ≥90%), uncertain significance (10–90%), likely benign (LB; ≤10%), and benign (≤1%). Classification combines independent lines of evidence, each characterised by a direction (pathogenic or benign) and strength (supporting, moderate, strong, or very strong), according to fixed combination rules [1]. Tavtigian et al. [2, 3] formalised this as a Bayesian point system in which each evidence strength carries an integer point value (±1, ±2, ±4, ±8) proportional to $\log(\mathrm{LR}^+)$ (the natural logarithm of the likelihood ratio), so that the combined LR^+ for total points T is $\mathrm{LR}^+(T) = C^{T/8}$ for a single constant $C = O_{\mathrm{PVSt}}$.

Evidence calibration pipelines—including our own ExCALIBR [7]—convert continuous assay scores to evidence points by comparing the variant's score-specific LR^+ to the thresholds $C^{k/8}$ for each evidence strength $k \in \{1, 2, 4, 8\}$. Supporting evidence requires $\mathrm{LR}^+(s) \geq C^{1/8}$; moderate requires $\mathrm{LR}^+(s) \geq C^{1/4}$; and so on [5]. These thresholds are correct only if the posterior probability at each boundary satisfies the ACMG target—that is, only if the correct C is used for the gene-specific prior. However, C^* in the Tavtigian framework is a single integer determined by grid search at the canonical prior $p = 0.10$, and applying it at other priors yields thresholds that map assay scores to incorrect posterior probabilities.

Table 1: Tavtigian boundary posteriors vs. piecewise targets at five representative priors. Piecewise values are exact at each target by construction. C^* is from the integer grid search.

p	C^*	LP (6 pts, tgt 0.90)		P (10 pts, tgt 0.99)		LB (-1 pts, tgt 0.10)		B (-7 pts, tgt 0.01)	
		Tav	PW	Tav	PW	Tav	PW	Tav	PW
0.01	8511	**.900**	.900	.999	**.990**	.003	**.100**	.000	**.010**
0.05	950	**.900**	.900	.996	**.990**	.022	**.100**	.000	**.010**
0.10	350	**.900**	.900	.994	**.990**	.051	**.100**	.001	**.010**
0.25	92	.908	**.900**	**.990**	.990	.159	**.100**	.006	**.010**
0.45	2876	.997	**.900**	1.00	**.990**	.232	**.100**	.001	**.010**

Harrison et al. [6] observed that variants classified as LP in ClinVar often carry posterior probabilities well below the nominal 90%, and that reclassification data reveal systematic inconsistency in posterior calibration across genes. A key source of this inconsistency is the prior-dependence of the correct classification threshold: a variant whose LR^+ just reaches the LP threshold at $p = 0.10$ would yield a posterior of only 0.45 at $p = 0.01$—squarely in the VUS range—while the same threshold yields 0.96 at $p = 0.25$. Functional assay calibration pipelines that apply Tavtigian thresholds without prior-adaptation therefore produce evidence assignments whose implied posteriors depend on the gene-specific prior in an uncontrolled way. This is not a limitation of the underlying assay or calibration model, but of the threshold mapping from LR^+ to evidence strength.

Here, we characterise the calibration errors introduced by prior-fixed Tavtigian thresholds, prove that this limitation is unavoidable for any additive system with prior-independent thresholds, and describe a piecewise-linear Bayesian solution that achieves exact boundary posteriors at every prior. We provide quantitative comparisons of evidence threshold accuracy across the biologically relevant prior range and discuss the implications for assay calibration pipelines such as ExCALIBR [7], OddsPath [4], and PP3/BP4 calibration [5].

The Tavtigian Framework and Its Prior-Dependent Failures

Construction and calibration at the canonical prior

Given a prior p and a combined $\mathrm{LR}^+ = C^{T/8}$, the posterior probability of pathogenicity is

$$P(Y{=}1 \mid E{=}e) = \frac{C^{T/8}\, p}{(C^{T/8} - 1)p + 1}. \tag{1}$$

The constant C (the "OddsPath" O_{PVSt}) is determined from the 2020 derivation [3]: five of the six LP combination rules each require odds equivalent to six pieces of supporting evidence, fixing $O_{\mathrm{PSu}}^6 = 81$, hence $C^* = O_{\mathrm{PSu}}^8 = 81^{4/3} \approx 350$ at $p = 0.10$. To resolve two originally inconsistent ACMG rules, we follow the modification of Tavtigian et al. [2]: the combination 2S is moved to the LP classification (8 points, Post_P ≈ 0.975) and VS+M is moved to the P classification (10 points, Post_P ≈ 0.994); all remaining rules are internally consistent at $p = 0.10$ with $C^* = 350$.

Threshold accuracy degrades away from $p = 0.10$

Table 1 and Figure 2 show the posteriors produced by Tavtigian thresholds at the four ACMG classification boundaries across the prior range. Three distinct failure modes are evident.

Pathogenic classification ranges: accurate near $p = 0.10$, drifting elsewhere. The LP points cutoff (6 points) posterior is within 1% of the 0.90 target for priors between 0.01 and 0.25, because the LP constraint is the binding one in the grid search at low priors and C^* approximately tracks the LP-optimal value. Above $p \approx 0.25$, however, benign combination rules become binding and drive C^* to anomalously large values: at $p = 0.45$, the grid search returns $C^* = 2876$, causing the LP and P points cutoffs to reach posteriors of 0.997 and ≈1.000, respectively—variants are classified with far greater confidence than warranted. The P points cutoff is persistently below its 0.99 target across all priors, ranging from 0.990 to 0.999.

Benign classification ranges: severely miscalibrated across the full range. The LB points cutoff (−1 points, target: posterior ≤0.10) fails at every prior in Table 1 except $p = 0.25$. At $p = 0.01$, the LB points cutoff posterior is 0.003—the Tavtigian LR threshold is 33× more stringent than necessary. A gene with prior $p = 0.01$ would require a variant's LR^+ to fall below 0.26 to receive Supporting Benign evidence, when the correct LR threshold (achieving the target posterior of 0.10) is $\mathrm{LR}^+ < 11$. Equivalently, every variant with $0.26 < \mathrm{LR}^+ < 11$ that should be classified as LB is instead returned as VUS. At $p = 0.25$, the failure inverts: the LB posterior is 0.159, so variants are classified LB when their true posterior is still 15.9%, above the intended threshold.

The implication for assay calibration. When a calibration pipeline (e.g., ExCALIBR, OddsPath, PP3/BP4) uses $C^{k/8}$ to assign evidence strengths, the resulting assignments imply the ACMG posterior targets only at $p = 0.10$. For any other prior—including the gnomAD-estimated priors that ExCALIBR computes per gene [7]—the evidence strengths carry the wrong implied posteriors. Benign evidence is particularly affected: at low priors (gene-level rarity), the Tavtigian LB threshold is far too stringent, suppressing benign evidence assignments for variants that genuinely meet the LB posterior criterion. This provides a mechanistic explanation for the observation that LP classifications in ClinVar frequently carry posteriors well below 90% [6].

A Mathematical Trilemma

The miscalibration above is not a limitation of C^* estimation but a provable consequence of requiring three mutually incompatible properties.

Theorem 1. *No scoring system satisfying fixed combination thresholds* (R1)*, the* 1:2:4:8 *tier ratio* (R2)*, and additive evidence combination* (R3) *can simultaneously achieve exact classification boundary posteriors at all priors* (R4)*.*

Proof. R2–R3 require $\log(\mathrm{LR}^+(T)) = \alpha T$ for a prior-independent constant α (i.e., a linear relationship between $\log(\mathrm{LR}^+)$ and T). R4 at the LP boundary then requires $e^{\alpha T_{LP}} = 9(1-p)/p$, which is strictly decreasing in p while the left side is constant. The equation holds for at most one p, contradicting R4. □

Corollary 1. *The same argument applies to all four classification boundaries, and any rescaling of the tier weights preserves the contradiction.*

The trilemma—Table 2—leaves three design choices. Tavtigian [2] satisfies R1–R3 (additive, fixed thresholds) at the cost of R4 (posteriors drift with prior). A "floating threshold" approach satisfies R3–R4 but breaks R1, since the same evidence combination would cross different classification boundaries at different priors—unacceptable when rule-based evidence sources already incorporate the prior independently. The piecewise method (below) satisfies R1, R2, and R4 exactly, sacrificing only strict additivity (R3) in a quantifiable way.

Table 2: The trilemma: any two of R1, R3, R4 can be satisfied exactly, but not all three.

Method	R1 (fixed thresh.)	R2 (1:2:4:8)	R3 (additive)	R4 (exact post.)
Tavtigian [2]	✓	✓	✓	×
Floating threshold	×	✓	✓	✓
Piecewise	✓	✓	≈	✓

A Piecewise Bayesian Solution

Construction and exact boundary posteriors

At each prior p, the four Bayesian target LR^+ values are computed analytically [8]:

$$\mathrm{LR^+}{}^*_q(p) = \frac{q(1-p)}{p(1-q)}, \qquad q \in \{0.99,\, 0.90,\, 0.10,\, 0.01\}, \tag{2}$$

yielding $P(Y{=}1 \mid E{=}e) = q$ exactly. These are anchored to the ACMG classification boundaries $T \in \{10, 6, -1, -7\}$ to define four knots of a piecewise-linear function $L_p(T) = \log(\mathrm{LR}^+(T))$ (the natural logarithm of the likelihood ratio), with linear interpolation within each of the three segments and extrapolation beyond the outermost knots using the adjacent segment's slope. The posterior is then given by the logistic function applied to $L_p(T)$ plus the log-prior-odds: $P(Y{=}1 \mid E{=}e) = \sigma\big(L_p(T) + \log \frac{p}{1-p}\big)$, where σ is the logistic function and $\log \frac{p}{1-p}$ is the log-prior-odds. By construction, $P(Y{=}1 \mid E{=}e) = q$ exactly at each of the four points cutoff values for every prior $p \in (0, 1)$.

A key property is that the slopes of $\log(\mathrm{LR}^+)$ vs. T in each segment are prior-independent: $\alpha_{\text{B–LB}} = \log(11)/6 \approx 0.385$ per point, $\alpha_{\text{LB–LP}} = \log(81)/7 \approx 0.586$ per point, and $\alpha_{\text{LP–P}} = \log(11)/4 \approx 0.614$ per point. Only the vertical position (intercept) of the $\log(\mathrm{LR}^+)$ line shifts with prior, ensuring the correct anchor LR^+ (the threshold LR^+ value) at every classification boundary. Figure 3 illustrates this geometry: as p changes, the piecewise line translates vertically through all four Bayesian target dots, while the Tavtigian straight line rotates and misses the dots at priors away from 0.10.

Evidence thresholds are substantially different from Tavtigian at non-canonical priors

The practical consequence of piecewise calibration is most visible in the evidence thresholds themselves. At the LP points cutoff (6 points), both methods agree by construction at $p = 0.10$: $\mathrm{LR}^{+*} = 81$. But for the LB points cutoff (-1 points):

$$\mathrm{LR^+}{}^*_{\text{PW}}(-1, p) = \frac{0.10(1-p)}{p \cdot 0.90}, \qquad \mathrm{LR^+}{}^*_{\text{Tav}}(-1, p) = C^*(p)^{-1/8}, \tag{3}$$

and these diverge sharply. At $p = 0.01$: $\mathrm{LR^+}{}^*_{\text{PW}} = 11.0$, $\mathrm{LR^+}{}^*_{\text{Tav}} = 0.26$. Any variant with $0.26 < \mathrm{LR}^+ < 11.0$—a 42-fold range in LR^+ (or equivalently $\log(11.0/0.26) \approx 1.62$ in $\log(\mathrm{LR}^+)$)—receives benign evidence under piecewise but not under Tavtigian. For the LP points cutoff at $p = 0.45$: $\mathrm{LR^+}{}^*_{\text{PW}} = 9.2$, $\mathrm{LR^+}{}^*_{\text{Tav}} = 2876^{0.75} \approx 477$—Tavtigian requires 52× more evidence to reach LP classification than necessary.

These threshold differences translate directly into evidence assignment errors in calibration pipelines. A calibration tool that estimates a variant's LR^+ correctly but maps it to evidence strengths using Tavtigian thresholds will under-assign benign evidence at low priors and over-assign pathogenic evidence at high priors.

Combination Errors: a Secondary Consideration

When two pieces of evidence are combined by summing points, the combined LR^+ should equal the product of individual LR^+ values. Tavtigian satisfies this exactly by construction ($C^{a+b} = C^a \cdot C^b$), but its per-code LR^+ thresholds are miscalibrated away from $p = 0.10$. The piecewise method has exact single-assay boundary posteriors but incurs combination errors when evidence sums cross a segment boundary (Figure 4).

Piecewise combination errors arise from the fact that the $\log(\mathrm{LR}^+)$ function is anchored at the LB points cutoff (-1 points), not at the origin (0 points), so the piecewise segments have the form $L_p(T) = L_p(-1) + \alpha(T - (-1))$ rather than αT. Within a single segment, this produces a constant offset error in $\log(\mathrm{LR}^+)$ per combined pair. The 6·11·6 additive variant of Tavtigian [3] is specifically designed to eliminate within-segment errors by moving the LB points cutoff to 0 points.

We measure combination errors using the absolute log-odds metric $|\mathrm{logit}(\hat{p}) - \mathrm{logit}(p_{\mathrm{ref}})|$, where p_{ref} is the Bayesian posterior from the piecewise per-code LR^+ product. This metric is scale-invariant and treats errors near 0 and 1 appropriately—a posterior shift from 0.990 to 0.999 is as significant as one from 0.010 to 0.019, both corresponding to roughly 0.9 log-odds units.

Table 3 shows median log-odds errors across the full prior range. For mixed combinations, piecewise wins consistently and significantly at every prior ($p < 0.01$, Wilcoxon signed-rank; Table **??**). For within-segment pathogenic pairs, piecewise also wins at all priors, reaching significance only at $p = 0.45$ where Tavtigian's C^* severely degrades. The one consistent exception is cross-segment evidence: Tavtigian wins at $p \leq 0.25$ due to its exact additivity by construction, though piecewise reverses this at $p = 0.45$.

The primary advantage of piecewise is therefore in single-assay LR^+ threshold correctness, not combination accuracy, which is the property that directly impacts assay calibration pipelines.

Discussion

Prior-adaptive thresholds are necessary for accurate evidence assignment

Assay calibration pipelines estimate the variant-level positive likelihood ratio from data; the quality of this estimation is the primary driver of calibration accuracy [5, 7]. However, once $\mathrm{LR}^+(s)$ is estimated, the mapping to evidence strengths introduces a second source of error: the use of prior-fixed thresholds. Our results show that this error is largest in the benign direction at low priors, which is precisely the regime most clinically relevant for rare disease panels where gene-level priors can be well below 0.05. The piecewise method eliminates this second source of error entirely by constructing prior-adaptive thresholds that guarantee the ACMG posterior targets at every prior.

Implications for assay calibration with gene-specific priors

ExCALIBR [7] and related methods estimate the gene-specific prior from gnomAD population frequencies, which vary substantially across genes. When these priors are used to determine C^* via the Tavtigian formula, the pathogenic LP and P thresholds are approximately correct (within $\sim$1% for LP at priors below 0.25), but the benign LB and B thresholds can be orders of magnitude too stringent. Replacing $C^{k/8}$ thresholds with piecewise LR^+ thresholds in these pipelines would recover the correct ACMG boundary posteriors and substantially increase benign evidence assignment rates at low priors. At $p = 0.01$—representative of high-throughput panels screening many genes simultaneously—every variant with $0.26 < \mathrm{LR}^+ < 11$ would gain Supporting Benign evidence under piecewise; at Supporting Benign, the posterior probability is by construction ≤ 0.10.

Combination errors and multi-assay integration

For applications combining multiple independent assays or combining assay evidence with other evidence types (computational, segregation, population), the additivity property of the evidence combination is relevant. Tavtigian's exact additivity ($C^{a+b} = C^a \cdot C^b$) is convenient but does not compensate for the threshold miscalibration shown above; a correctly additive system with the wrong thresholds still produces wrong posteriors. Piecewise errors of $\sim$0.005–0.06 in posterior probability at the cross-segment or within-segment level are small relative to the 42-fold difference in LB LR threshold at $p = 0.01$. When multiple assay results are combined with other lines of evidence, the combination errors of the piecewise method are bounded by the maximum log-slope discontinuity across segment boundaries, which is a fixed geometric property independent of prior.

Conclusion

We prove that the Tavtigian framework's threshold miscalibration at non-canonical priors is an unavoidable mathematical consequence of requiring additive evidence combination with fixed, prior-independent thresholds. The piecewise solution resolves this by sacrificing strict additivity in exchange for exact boundary posteriors at every prior, preserving the ACMG combination rules and point system. The practical impact is largest for benign evidence at low priors, where Tavtigian thresholds are far too stringent and cause widespread under-assignment of benign evidence. We recommend that assay calibration pipelines that estimate gene-specific priors use prior-adaptive (piecewise) thresholds in the LR^+-to-evidence-strength mapping step.

Table 3: Median absolute log-odds combination error by type across key priors (lower is better, bolded). Errors measured as $|\text{logit}(\hat{p}) - \text{logit}(p_{\text{ref}})|$.

Type	$p = 0.01$		$p = 0.05$		$p = 0.10$		$p = 0.25$		$p = 0.45$	
	PW	**Tav**	**PW**	**Tav**	**PW**	**Tav**	**PW**	**Tav**	**PW**	**Tav**
Within-seg (path)	**3.03**	3.28	**1.38**	1.49	**0.628**	0.681	**0.471**	0.598	**1.37**	4.76
Cross-seg (path)	3.11	**1.04**	1.46	**0.487**	0.713	**0.235**	0.386	**0.359**	**1.28**	6.42
Within-seg (ben)	**2.80**	8.52	**1.15**	4.12	**0.400**	2.13	**0.699**	0.736	1.60	**0.810**
Cross-seg (ben)	**2.80**	12.9	**1.15**	6.87	**0.400**	4.13	0.699	**0.258**	1.60	**2.77**
Mixed (path+ben)	**3.25**	6.51	**1.60**	3.09	**0.856**	1.54	**0.243**	0.503	**1.14**	2.33

Table 4: Wilcoxon signed-rank test on per-combination log-odds error differences (PW − Tav). Null hypothesis: median difference = 0. W = test statistic; p = two-sided p-value. Bold p-values indicate significance at $\alpha = 0.05$. Note: within/cross benign types have only 5 combinations; minimum attainable $p = 0.063$.

Type	$p = 0.01$		$p = 0.05$		$p = 0.10$		$p = 0.25$		$p = 0.45$	
	W	p	W	p	W	p	W	p	W	p
Within-seg (path)	30	0.52	36	0.85	30	0.52	23	0.23	0	**0.001**
Cross-seg (path)	0	**<0.001**	0	**<0.001**	0	**<0.001**	68	**0.018**	0	**<0.001**
Within-seg (ben)	0	0.063	0	0.063	0	0.063	6	0.81	2	0.19
Cross-seg (ben)	0	0.063	0	0.063	0	0.063	3	0.31	1	0.13
Mixed (path+ben)	2	**<0.001**	3	**<0.001**	2	**<0.001**	23	**0.018**	4	**<0.001**

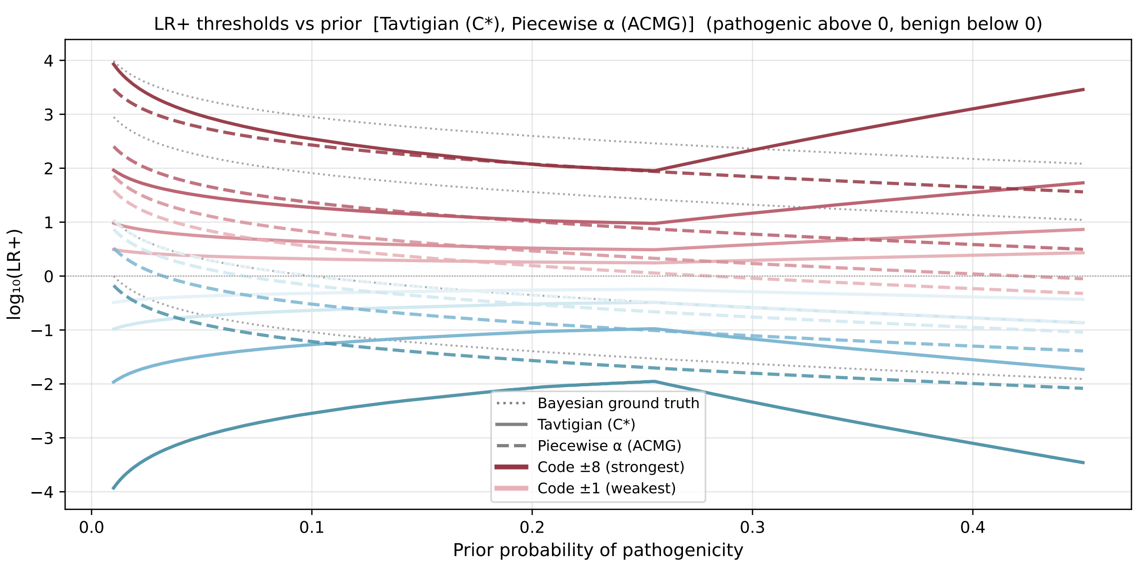

Figure 1: LR^+ thresholds for evidence codes 1, 2, 4, 8 (Su, M, S, VS) vs. gene-specific prior. Tavtigian (solid): follows $C^{*T/8}$; shows a non-monotone artefact near $p \approx 0.25$ where C^* passes through its minimum, and a large spike at $p = 0.45$ where benign combination rules drive C^* to 2876. Piecewise (dashed): varies smoothly; thresholds are prior-adaptive and produce exact posteriors at each evidence boundary. Dotted lines: analytical Bayesian thresholds (continuous ground truth). Warm colours = pathogenic codes; cool colours = benign codes.

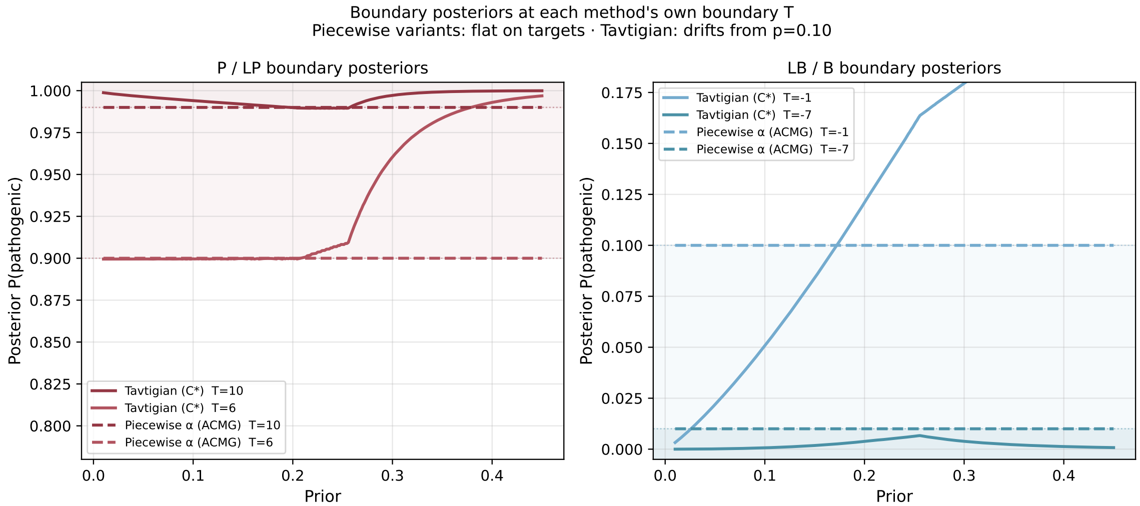

Figure 2: Posterior probability at each method's classification boundary T value vs. gene-specific prior. Piecewise (dashed): flat at all four targets by construction. Tavtigian (solid): well-calibrated for LP at priors ≤ 0.25 but severely miscalibrated for LB and B boundaries across the full prior range; anomalous over-calibration at $p \approx 0.45$.

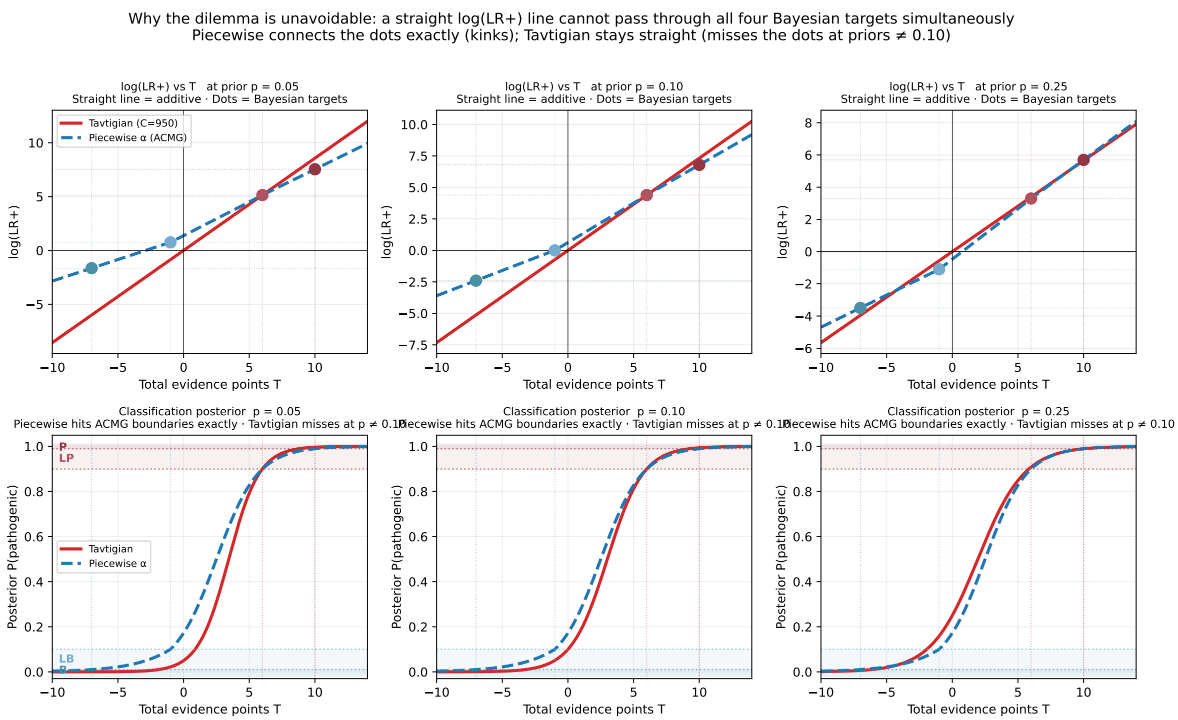

Figure 3: $\log \mathrm{LR}^+$ vs. total evidence points T at three prior values. Coloured dots indicate the exact Bayesian target $\log \mathrm{LR}^+$ at each of the four classification boundaries. Tavtigian (red): a single straight line that passes through all four dots only at $p = 0.10$. Piecewise (blue): a kinked line that passes through all four dots at every prior by translating vertically. Lower panels: the corresponding posterior probability curves; piecewise reaches each ACMG boundary exactly, Tavtigian drifts.

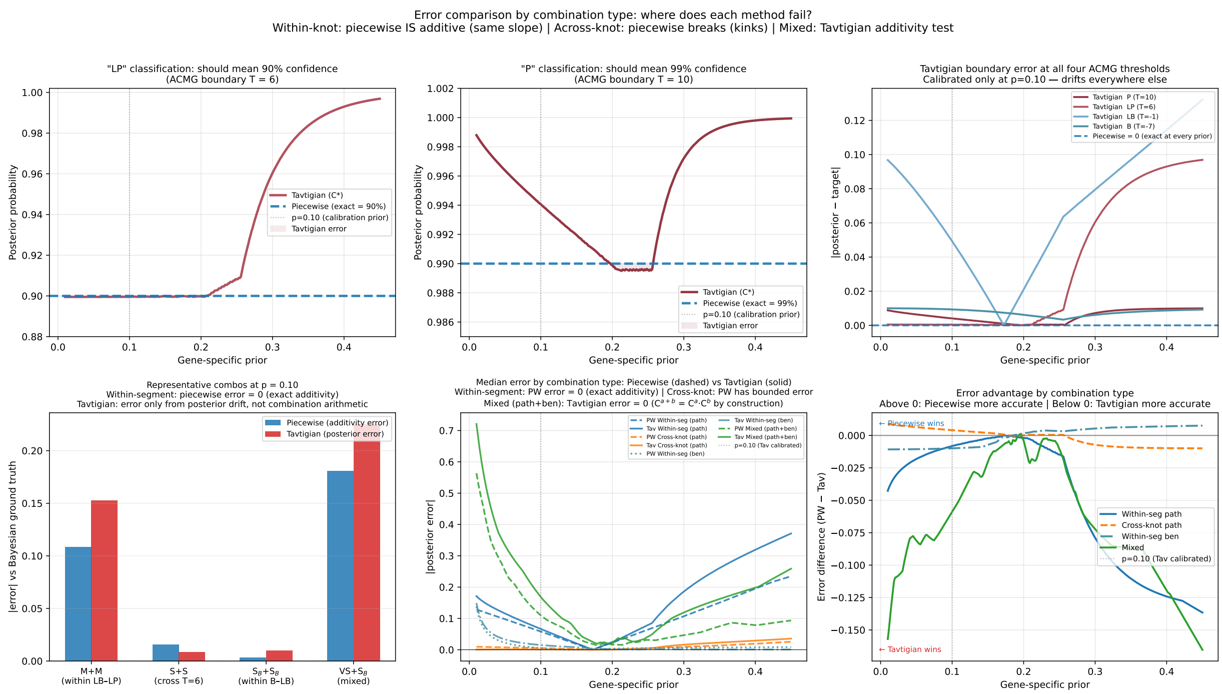

Figure 4: Evidence combination errors stratified by type. *Top row*: single-assay boundary posteriors showing Tavtigian drift and piecewise exactness. *Bottom left*: representative per-type errors at $p = 0.10$; both methods measured against the same Bayesian ground truth. *Bottom middle*: median per-type combination errors vs. prior (piecewise dashed, Tavtigian solid). *Bottom right*: error advantage (piecewise − Tavtigian); positive = piecewise more accurate.

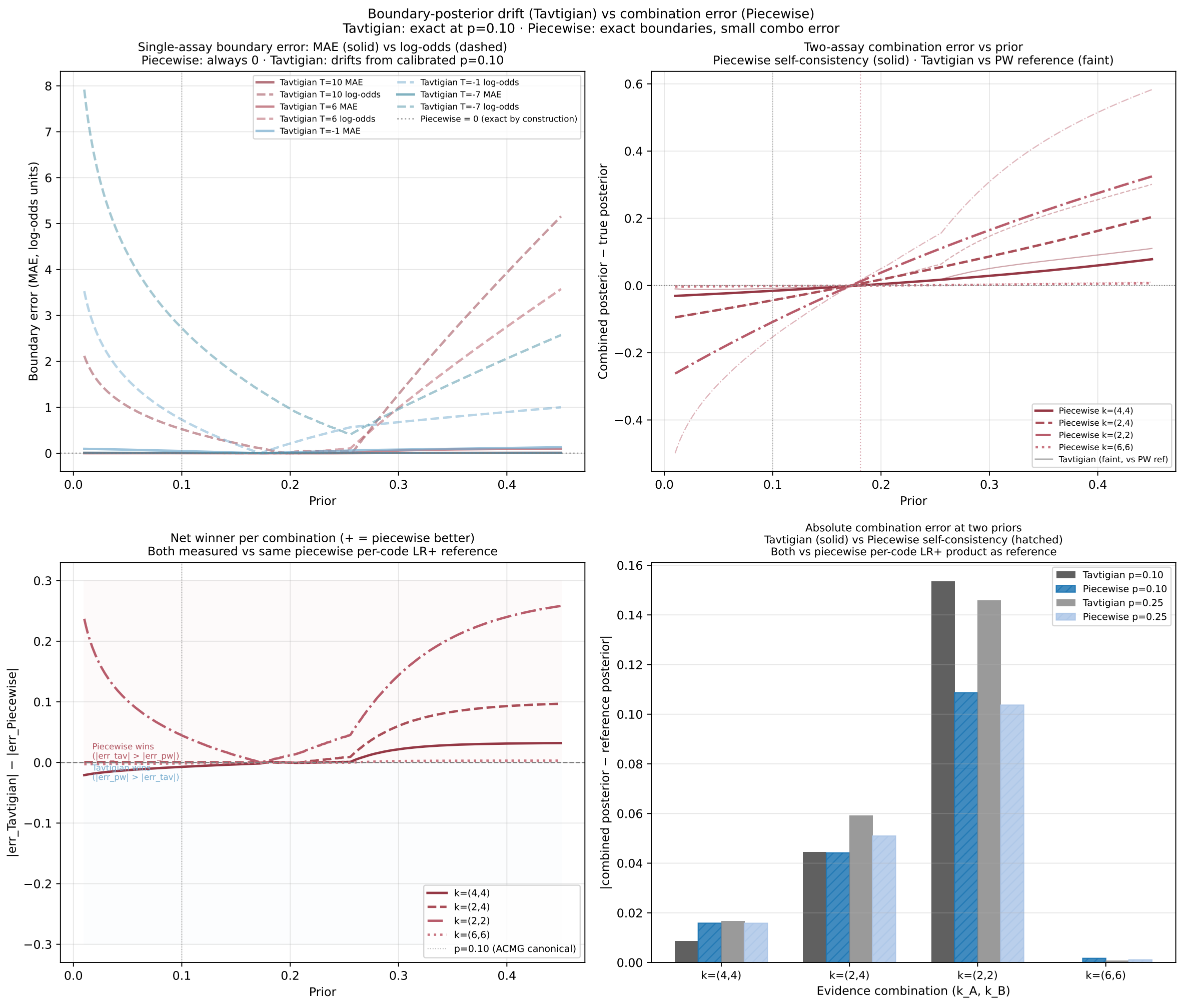

Figure 5: Single-assay boundary errors for both methods using two error metrics: MAE in posterior probability (solid) and absolute log-odds error $|\text{logit}(\hat{p}) - \text{logit}(\text{target})|$ (dashed). Piecewise $= 0$ at all boundaries by construction. The log-odds metric reveals the true severity of Tavtigian's benign-boundary miscalibration at low priors, which MAE substantially understates.

Supplementary: evidence tier point system and combination rules

Table 5: ACMG/AMP combination rules in the modified Tavtigian point system [2, 3]. Points: Su = 1, M = 2, S = 4, VS = 8; $\mathrm{Su}_B = -1$, $\mathrm{S}_B = -4$ (standard ACMG benign tiers only). †Modified rule: original ACMG classifies VS+M as LP and 2S as P; both are internally inconsistent at $p = 0.10$, so following Tavtigian et al. they are swapped. All calculations use the modified (swapped) rules.

Category	Combination	Points
P	VS + S	12
P	VS + 2M	12
P	VS + M + Su	11
P	VS + M †	10
P	S + 3M	10
P	S + 2M + 2Su	10
P	S + M + 4Su	10
LP	S + M	6
LP	S + 2Su	6
LP	3M	6
LP	2M + 2Su	6
LP	M + 4Su	6
LP	2S †	8
LB	S_B + Su_B	−5
LB	$2\mathrm{Su}_B$	−2
B	$2\mathrm{S}_B$	−8

Supplementary: combination error tables

The following tables list the posterior combination errors for all evidence pairs at priors $p \in \{0.01, 0.10, 0.25, 0.45\}$. PW = piecewise additivity error; Tav = Tavtigian posterior error, both measured against the Bayesian posterior from the piecewise per-code LR^+ product. Standard ACMG pathogenic tiers: Su= 1, M= 2, S= 4, VS= 8; codes 3,5,6,7 are intermediate strengths not corresponding to named tiers. Standard ACMG benign tiers: $\mathrm{Su}_B = -1$, $\mathrm{S}_B = -4$; M_B (−2, †) and VS_B (−8, †) are non-standard hypothetical extensions.

Table 6: Combination errors for all pathogenic code pairs. $k_A \le k_B$, both in $\{1, 2, 3, 4, 5, 6, 7, 8\}$. Codes 3,5,6,7 are intermediate strengths not corresponding to named ACMG tiers; codes 1,2,4,8 = Su, M, S, VS. PW = piecewise additivity error; Tav = Tavtigian posterior error. Both measured vs. the Bayesian posterior from the piecewise per-code LR^+ product.

k_A	k_B	T	$p = 0.01$		$p = 0.1$		$p = 0.25$		$p = 0.45$	
			PW	**Tav**	**PW**	**Tav**	**PW**	**Tav**	**PW**	**Tav**
Su	Su	2	**3.03**	5.04	**0.628**	1.05	**0.471**	0.817	**1.37**	3.47

continued. . .

k_A	k_B	T	$p = 0.01$		$p = 0.1$		$p = 0.25$		$p = 0.45$	
			PW	**Tav**	**PW**	**Tav**	**PW**	**Tav**	**PW**	**Tav**
Su	M	3	**3.03**	4.54	**0.628**	0.942	**0.471**	0.754	**1.37**	3.84
Su	Su+M?	4	**3.03**	4.04	**0.628**	0.838	**0.471**	0.691	**1.37**	4.21
Su	S	5	**3.03**	3.53	**0.628**	0.733	**0.471**	0.629	**1.37**	4.58
Su	Su+S?	6	**3.03**	3.03	**0.628**	0.629	**0.471**	0.566	**1.37**	4.94
Su	M+S?	7	3.05	**2.53**	0.656	**0.524**	**0.443**	0.504	**1.34**	5.31
Su	Su+M+S?	8	3.05	**2.00**	0.656	**0.392**	**0.443**	0.470	**1.34**	5.71
Su	VS	9	3.05	**1.46**	0.656	**0.259**	0.443	**0.435**	**1.34**	6.10
M	M	4	**3.03**	4.04	**0.628**	0.838	**0.471**	0.691	**1.37**	4.21
M	Su+M?	5	**3.03**	3.53	**0.628**	0.733	**0.471**	0.629	**1.37**	4.58
M	S	6	**3.03**	3.03	**0.628**	0.629	**0.471**	0.566	**1.37**	4.94
M	Su+S?	7	3.05	**2.53**	0.656	**0.524**	**0.443**	0.504	**1.34**	5.31
M	M+S?	8	3.08	**2.02**	0.684	**0.420**	**0.414**	0.441	**1.31**	5.68
M	Su+M+S?	9	3.08	**1.49**	0.684	**0.287**	0.414	**0.407**	**1.31**	6.08
M	VS	10	3.08	**0.961**	0.684	**0.154**	0.414	**0.373**	**1.31**	6.47
Su+M?	Su+M?	6	**3.03**	3.03	**0.628**	0.629	**0.471**	0.566	**1.37**	4.94
Su+M?	S	7	3.05	**2.53**	0.656	**0.524**	**0.443**	0.504	**1.34**	5.31
Su+M?	Su+S?	8	3.08	**2.02**	0.684	**0.420**	**0.414**	0.441	**1.31**	5.68
Su+M?	M+S?	9	3.11	**1.52**	0.713	**0.315**	0.386	**0.379**	**1.28**	6.05
Su+M?	Su+M+S?	10	3.11	**0.989**	0.713	**0.183**	0.386	**0.344**	**1.28**	6.44
Su+M?	VS	11	3.11	**0.458**	0.713	**0.050**	0.386	**0.310**	**1.28**	6.84
S	S	8	3.08	**2.02**	0.684	**0.420**	**0.414**	0.441	**1.31**	5.68
S	Su+S?	9	3.11	**1.52**	0.713	**0.315**	0.386	**0.379**	**1.28**	6.05
S	M+S?	10	3.14	**1.02**	0.741	**0.211**	0.358	**0.316**	**1.26**	6.41
S	Su+M+S?	11	3.14	**0.486**	0.741	**0.078**	0.358	**0.282**	**1.26**	6.81
S	VS	12	3.14	**0.046**	0.741	**0.055**	0.358	**0.248**	**1.26**	7.21
Su+S?	Su+S?	10	3.14	**1.02**	0.741	**0.211**	0.358	**0.316**	**1.26**	6.41
Su+S?	M+S?	11	3.17	**0.514**	0.769	**0.106**	0.329	**0.254**	**1.23**	6.78
Su+S?	Su+M+S?	12	3.17	**0.017**	0.769	**0.026**	0.329	**0.219**	**1.23**	7.18
Su+S?	VS	13	3.17	**0.549**	0.769	**0.159**	0.329	**0.185**	**1.23**	7.57
M+S?	M+S?	12	3.20	**0.011**	0.798	**0.002**	0.301	**0.191**	**1.20**	7.15
M+S?	Su+M+S?	13	3.20	**0.521**	0.798	**0.131**	0.301	**0.157**	**1.20**	7.55
M+S?	VS	14	3.20	**1.05**	0.798	**0.264**	0.301	**0.123**	**1.20**	7.94
Su+M+S?	Su+M+S?	14	3.20	**1.05**	0.798	**0.264**	0.301	**0.123**	**1.20**	7.94
Su+M+S?	VS	15	3.20	**1.58**	0.798	**0.396**	0.301	**0.088**	**1.20**	8.34
VS	VS	16	3.20	**2.12**	0.798	**0.529**	0.301	**0.054**	**1.20**	8.73

Table 7: Combination errors for benign code pairs. Standard ACMG benign tiers are Su_B (-1 pt) and S_B (-4 pt); M_B (-2 pt, †) and VS_B (-8 pt, †) are non-standard hypothetical extensions included for completeness. Negative T totals indicate net benign evidence.

k_A	k_B	T	$p = 0.01$		$p = 0.1$		$p = 0.25$		$p = 0.45$	
			PW	**Tav**	**PW**	**Tav**	**PW**	**Tav**	**PW**	**Tav**
Su_B	Su_B	-2	**2.80**	7.06	**0.400**	1.46	**0.699**	1.07	**1.60**	2.00
Su_B	M_B	-3	**2.80**	7.79	**0.400**	1.80	**0.699**	0.901	1.60	**1.41**
Su_B	S_B	-5	**2.80**	9.25	**0.400**	2.46	0.699	**0.570**	1.60	**0.214**
Su_B	VS_B	-9	**2.80**	12.18	**0.400**	3.79	0.699	**0.092**	**1.60**	2.17
M_B	M_B	-4	**2.80**	8.52	**0.400**	2.13	**0.699**	0.736	1.60	**0.810**
M_B	S_B	-6	**2.80**	9.98	**0.400**	2.79	0.699	**0.404**	1.60	**0.381**
M_B	VS_B	-10	**2.80**	12.91	**0.400**	4.13	0.699	**0.258**	**1.60**	2.76
S_B	S_B	-8	**2.80**	11.45	**0.400**	3.46	0.699	**0.073**	1.60	**1.57**
S_B	VS_B	-12	**2.80**	14.37	**0.400**	4.79	0.699	**0.589**	**1.60**	3.96
VS_B	VS_B	-16	**2.80**	15.33	**0.400**	6.12	**0.699**	1.25	**1.60**	6.34

Table 8: Combination errors for mixed (pathogenic + benign) code pairs. $k_A > 0$ (pathogenic tier), $k_B < 0$ (benign tier). Tavtigian has zero internal combination error for mixed evidence by construction ($C^{a+b} = C^a \cdot C^b$), but its per-code LR^+ thresholds are miscalibrated at non-canonical priors, so the posterior errors are still large.

k_A	k_B	T	$p = 0.01$		$p = 0.1$		$p = 0.25$		$p = 0.45$	
			PW	**Tav**	**PW**	**Tav**	**PW**	**Tav**	**PW**	**Tav**
Su	Su_B	0	**3.03**	6.05	**0.628**	1.26	**0.471**	0.942	**1.37**	2.74
Su	M_B	-1	**3.25**	6.78	**0.856**	1.59	**0.243**	0.776	**1.14**	2.14
Su	S_B	-3	**3.25**	8.25	**0.856**	2.25	**0.243**	0.445	1.14	**0.950**
Su	VS_B	-7	**3.25**	11.17	**0.856**	3.58	0.243	**0.217**	**1.14**	1.43
M	Su_B	1	**3.03**	5.55	**0.628**	1.15	**0.471**	0.879	**1.37**	3.11
M	M_B	0	**3.25**	6.28	**0.856**	1.48	**0.243**	0.714	**1.14**	2.51
M	S_B	-2	**3.48**	7.74	**1.08**	2.15	**0.015**	0.382	**0.913**	1.32
M	VS_B	-6	**3.48**	10.67	**1.08**	3.48	**0.015**	0.280	**0.913**	1.07
S	Su_B	3	**3.03**	4.54	**0.628**	0.942	**0.471**	0.754	**1.37**	3.84
S	M_B	2	**3.25**	5.27	**0.856**	1.27	**0.243**	0.588	**1.14**	3.24
S	S_B	0	**3.71**	6.74	**1.31**	1.94	**0.214**	0.257	**0.684**	2.05
S	VS_B	-4	**3.94**	9.66	**1.54**	3.27	0.442	**0.405**	0.456	**0.330**
VS	Su_B	7	3.00	**2.47**	0.599	**0.468**	**0.499**	0.560	**1.40**	5.37
VS	M_B	6	**3.20**	3.20	**0.799**	0.800	**0.299**	0.395	**1.20**	4.77
VS	S_B	4	**3.65**	4.67	**1.26**	1.47	0.157	**0.064**	**0.741**	3.58
VS	VS_B	0	**4.57**	7.59	**2.17**	2.80	1.07	**0.599**	**0.172**	1.20

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